

1-8 INSTALLATION PROCEDURE

1-8-1 Installation Flow

Note

All on-site construction must be completed before equipment is delivered and installation starts. Attempting to install the system while construction is being completed will impact installation efficiency and further delay site completion. Making sure that all preinstallation and construction work is completed before equipment is delivered will usually result in an earlier turnover date.

The flow chart in Illustration 1-2 should be followed for an orderly and efficient installation of the Signa OpenSpeed system. Note that many procedures may be performed in parallel and may be performed in any order according to the specific situation of each site.

Note

The flow chart references the Tab number and the PIP (Pictorial Installation Procedure) Direction number for the procedure. PIP Directions are similar to cable maps in that they are to be removed from this binder and used at the location where hardware installation is being performed. The PIP Direction may be returned to the binder for future reference upon completion of the installation. Other Directions and installation drawings delivered with kits are also referenced by the flowchart..

This installation flowchart has been developed assuming that all Signa OpenSpeed equipment has been delivered together. Ensure that every part required is available before starting.

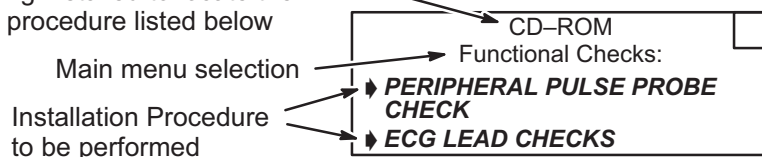
The general installation flow is as follows:

- Hardware installation per applicable Pictorial Installation Procedure (PIP) Direction
- Power-up and other pre-software load checks (this can be done while Magnet is being ramped)
- Software load from cold (this can be done while Magnet is being ramped)
- Function checks, Set-up and Calibration, and other activity requiring software running that can be done while magnet field is stabilizing
- Mechanical shim of Magnet
- System calibrations
- System performance checks
- Cover replacement
- Final sign-off and records
- Customer turn over.

1-8-2 Installation Flowchart Explanation

Shown below are three examples of the flow chart Installation Procedures and an explanation of the areas within the procedures.

Use the CD-ROM that applies to the System being installed to locate the installation procedure listed below

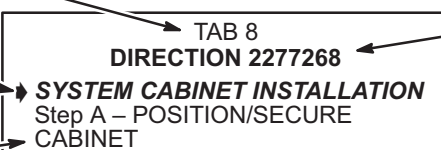


Use the correct Magnet Manual that applies to the System being installed to locate the installation procedure listed below



Tab within this manual (Direction 2196821) where the Direction listed below is located

Installation Procedure to be performed



Direction within the Tab where the Installation Procedure listed below is located

Specific steps or room of the above Installation Procedure to be performed at this time

LEGEND FOR FOLLOWING INSTALLATION FLOWCHART



INDICATES MILESTONE COMPLETION GOAL IN DAYS



INDICATES AND OPTIONAL PATH DEPENDING ON SYSTEM CONFIGURATION



INDICATES GE PROPRIETARY PROCEDURE



INDICATES AN ADDITIONAL PROPRIETARY TEST OR ALTERNATIVE PROPRIETARY CALIBRATION PROCEDURE AVAILABLE FOR GE USE OR TO CUSTOMERS WITH AN ADVANCED SERVICE PACKAGE LIMITED LICENSE.



AFTER GRADIENT SHIM (OR LV SHIM) IT MAY BE NECESSARY TO REPAIR THE MAGNET.

NOTE 1: STANDARD CORRELATED NOISE CHECK NOT REQUIRED IF PROPRIETARY SPT FULL TEST MODE PROCEDURE PERFORMED EARLIER DURING INSTALLATION (HOWEVER, MULTICOIL AND FAST RECEIVER OPTION CORRELATED NOISE CHECKS IN PROCEDURE STILL REQUIRED).

NOTE 2: IF TROUBLESHOOTING NECESSARY.

NOTE 3:

NOTE 4: A "saveINFO" SHOULD BE DONE BEFORE SPT IS RUN. IF YOU HAVE THE EPI OPTION, IT IS NECESSARY TO "saveINFO" AGAIN. IN THIS WAY, ANY DATA COLLECTED FOR CALIBRATION WILL NOT BE LOST IF A PROBLEM OCCURS.

NOTE 5: VALUES OF LVSHIM ARE DEPENDENT UPON EDDY CURRENT COMPENSATION.

NOTE 6: VALUES OF EDDY CURRENT COMPENSATION ARE INDEPENDENT OF MAGNET SHIM.

NOTE 7: OVERNIGHT DIAGS MAY BE RUN AT END OF ANY DAY TO DETECT POSSIBLE ERRORS.

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PERFORMED
ITEMS

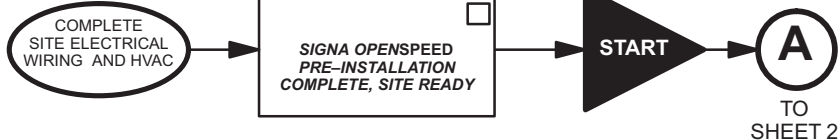
**IMPORTANT: PER ISO
9001 REQUIREMENTS**

NOTE:
Estimate *IS* AVERAGE MAN HOURS.

Estimate ## Hours	☐
Actual _____ Hours	☐

TASK TIME SPENT –
 THIS IS A DESIGN WITH SIX SIGMA
 PRODUCT (DFSS).
 WE NEED TO COLLECT DATA TO
 HELP ANALYZE THE PROCEDURES
 FOR IMPROVEMENT.
 PLEASE FILL IN THE ACTUAL
 HOURS SPENT ON A PROCEDURE.

PRE-INSTALLATION



*** MAGNET MANUAL USED:**

DIR 2243897 – GE 0.7T OpenSpeed Magnet Delivery/Installation Manual.
 DIR 2231934 – GE 0.7T OpenSpeed Magnet & Cryogenics Subsystem.

SIGNA OPENSPEED SYSTEM INSTALLATION FLOW CHART
 ILLUSTRATION 1–5 (SHEET 1 OF 12)

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**IMPORTANT: PER ISO
9001 REQUIREMENTS**

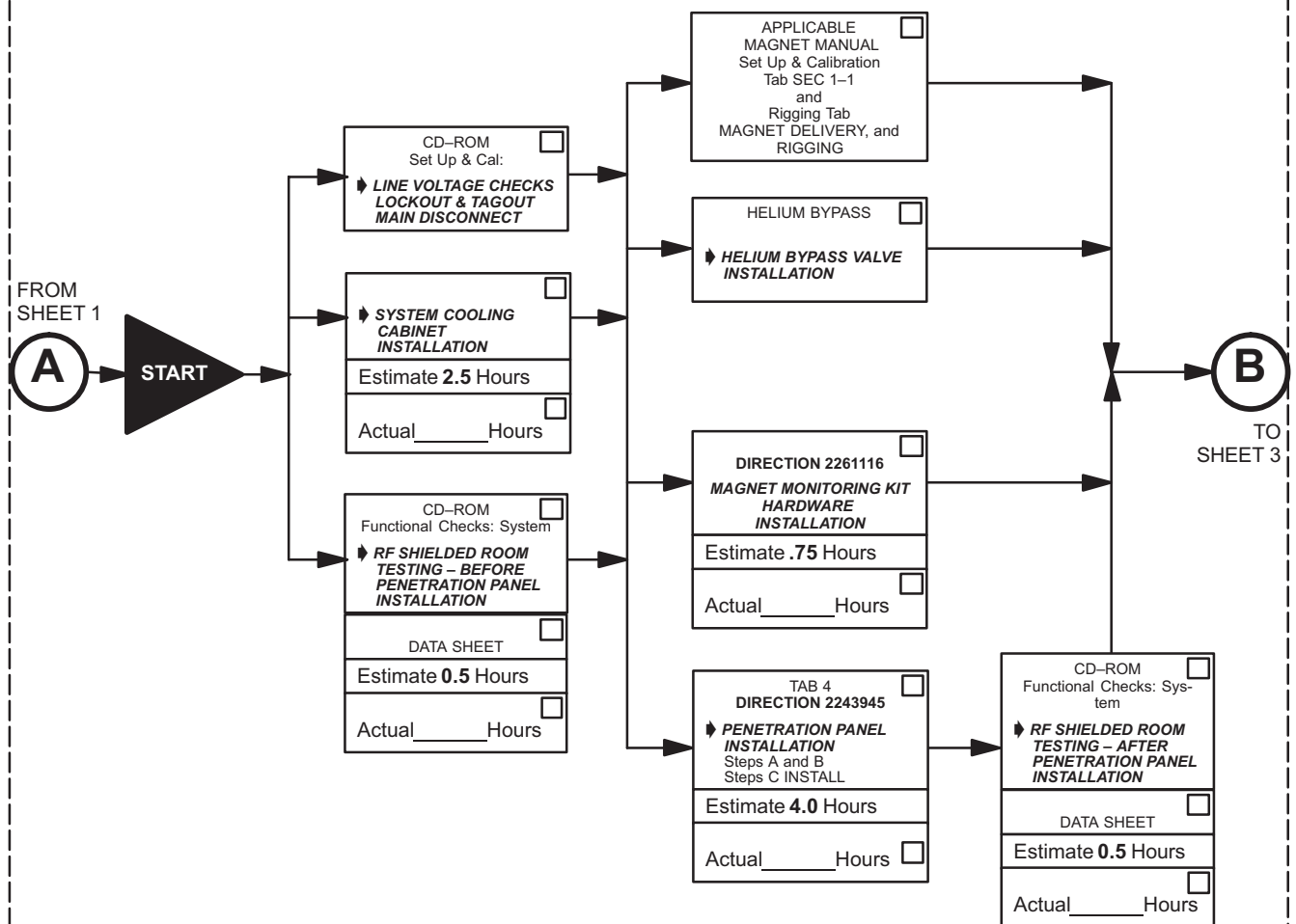
MAGNET MECHANICAL INSTALLATION

STEP 1

Estimated average elapsed
time: **8.0 hours**

Actual elapsed time: _____

Complete ☐



SIGNA OPENSPEED SYSTEM INSTALLATION FLOW CHART
ILLUSTRATION 1-5 (SHEET 2 OF 12)

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**IMPORTANT: PER ISO
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SYSTEM MECHANICAL INSTALLATION

STEP 2

Estimated average elapsed
time: **8.0 hours**

Actual elapsed time: _____

Complete ☐

FROM
SHEET 2



SIGNA OPENSPEED SYSTEM INSTALLATION FLOW CHART
ILLUSTRATION 1-5 (SHEET 3 OF 12)

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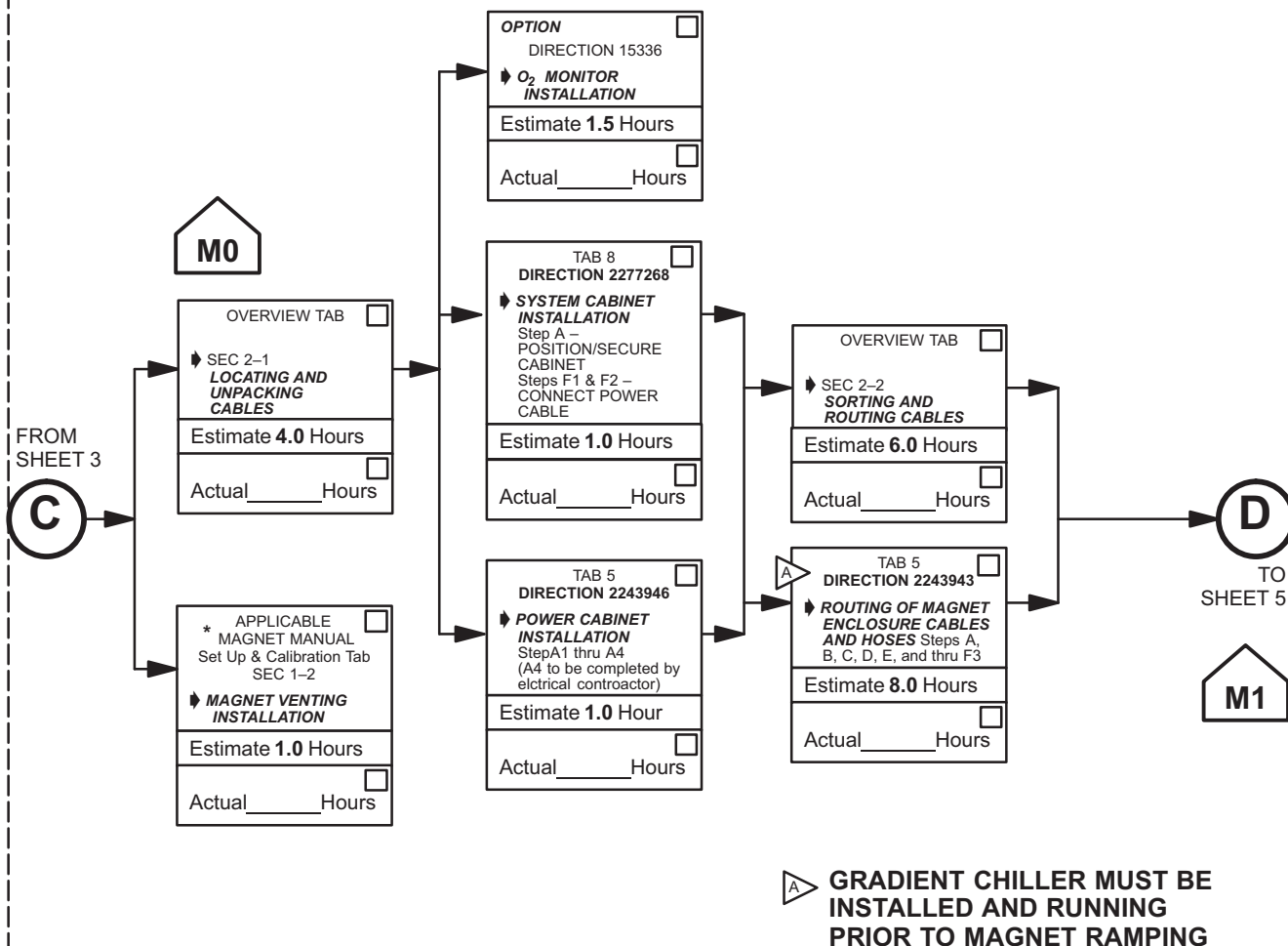
SYSTEM MECHANICAL INSTALLATION

STEP 3

Estimated average elapsed
time: **8.0 hours**

Actual elapsed time: _____

Complete ☐



SIGNA OPENSPEED SYSTEM INSTALLATION FLOW CHART
 ILLUSTRATION 1-5 (SHEET 4 OF 12)

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SYSTEM MECHANICAL INSTALLATION (CONTINUED)

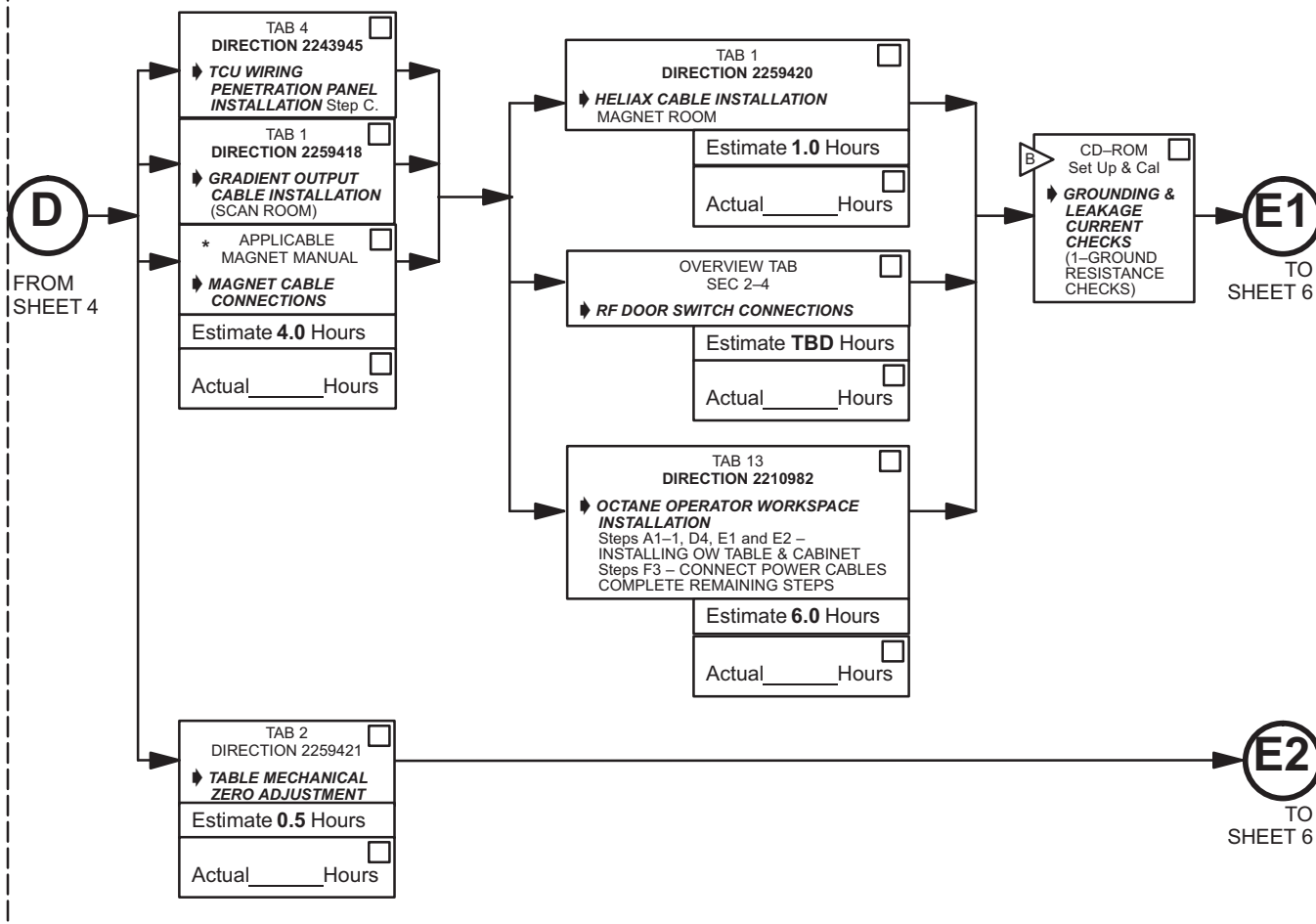
STEP 4

Estimated average elapsed
time: **4.0 hours**

Actual elapsed time: _____

Complete ☐

**▶ HARD WIRED SIGNAL CABLES ARE NOT TO BE
CONNECTED TO SUBSYSTEM CABINETS PRIOR
TO THIS CHECK. FIBER OPTIC CABLES MAY BE
CONNECTED AND NOT IMPACT THIS CHECK.**



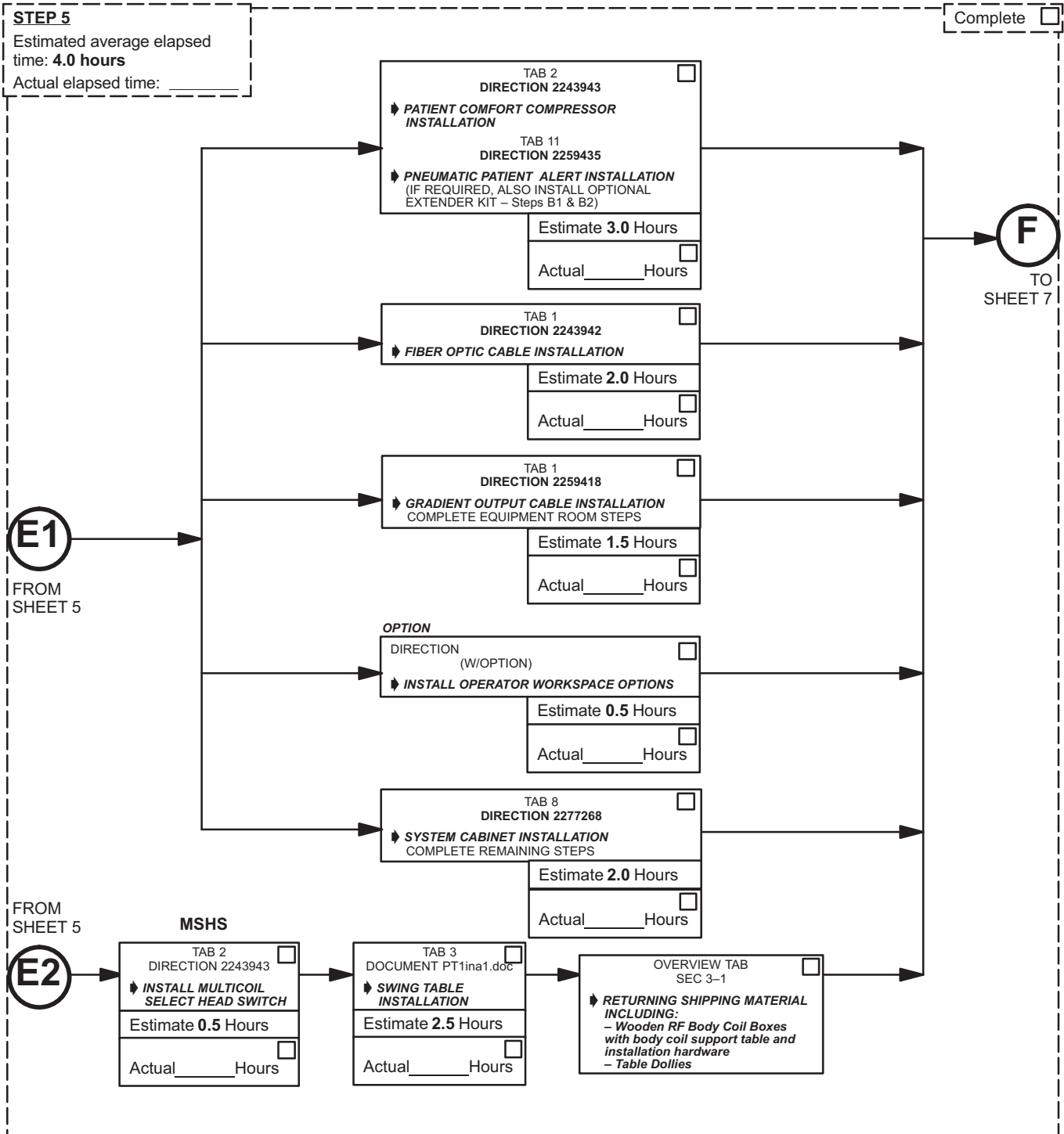
SIGNA OPENSPEED SYSTEM INSTALLATION FLOW CHART
ILLUSTRATION 1-5 (SHEET 5 OF 12)

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SYSTEM MECHANICAL INSTALLATION (CONTINUED)



SIGNA OPENSPEED SYSTEM INSTALLATION FLOW CHART
 ILLUSTRATION 1-5 (SHEET 6 OF 12)

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**IMPORTANT: PER ISO
9001 REQUIREMENTS**

SYSTEM PREPARATION FOR CALIBRATION

STEP 6

Estimated average elapsed
time: **2.0 hours**

Actual elapsed time: _____

Complete ☐

FROM
SHEET 6

F

IF NECESSARY,
RESOLVE SHIPMENT
SHORTAGES AND
OMISSIONS BY
MECHANICAL
SUPPLIER

CHECK MECHANICAL
FLOWCHART
TO INSURE ALL
STEPS ARE
COMPLETE

INSURE ALL RATING
PLATES ARE INSTALLED

RECHECK ALL WIRING
CONNECTIONS USING
CABLE MAP
CORRECT FOR WIRING
ERRORS IF NECESSARY

HANDOFF REVIEW WITH
MECHANICAL INSTALLATION
VENDOR

Estimate **TBD** Hours

Actual _____ Hours

READY FOR POWER UP

G
TO
SHEET 8

M2

SIGNA OPENSPEED SYSTEM INSTALLATION FLOW CHART
ILLUSTRATION 1-5 (SHEET 7 OF 12)

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**IMPORTANT: PER ISO
9001 REQUIREMENTS**

SYSTEM CALIBRATION

STEP 7

Estimated average elapsed
time: **28.0 hours**

Actual elapsed time: _____

**DO NOT TURN ON THE POWER CABINET CIRCUIT
BREAKERS (BREAKERS ON FRONT BOTTOM OF
CABINET) UNTIL ROOM HUMIDITY IS AT 60% OR LESS.**

**CAUTION! EQUIPMENT IS SUSCEPTABLE TO FAILURE
IN HIGH HUMIDITY.**

**FOR 8280 GRADIENT
AMPLIFIERS with ORM
ENTER:**
XFULL=28650 ± 405
YFULL=27650 ± 405
ZFULL=24650 ± 405

CD-ROM
System Procedures:
Functional Checks:
➔ **PDU E-STOP & OFF
CHECK**

OVERVIEW TAB
SEC 3-3
➔ **POWER ON**

**SYSTEM
POWER
ON**

FROM
SHEET 7

G

CD-ROM
Set Up & Cal:
➔ **ENVIRONMENTAL
CHECKS**

CD-ROM
Set Up/ Calibration
➔ **LOAD MAGNET
MONITOR SOFTWARE**
Estimate **0.5 Hours**
Actual _____ Hours

(SHOULD BE PRE-LOADED)

CD-ROM
Set Up/ Calibration
➔ **ENTITLE MAGNET
MONITOR**
Estimate **0.5 Hours**
Actual _____ Hours

* APPLICABLE
MAGNET MANUAL
Set Up & Calibration Tab SEC 3, 4
➔ **MAGNET FILL**
➔ **MAGNET PRE-RAMPING**
Set Up & Calibration Tab SEC 8 and 9
➔ **MAGNET RAMPING/FILLING**
➔ **HEATER CHECK**
Estimate **3.5 Hours**
Actual _____ Hours

M3
(NOTE 7)

* APPLICABLE
MAGNET MANUAL
Set Up & Calibration Tab
SEC 5
➔ **FIELD SETTLING**
CD-ROM
Set Up & Cal:
SHIM PROCEDURE
➔ **MAGNET DRIFT TEST**
Estimate **20.0 Hours**
Actual _____ Hours

TO
SHEET 9

H2

M4
(NOTE 7)

CD-ROM
Set Up/ Calibration
➔ **SAR CONFIG LIMITS**

TO
SHEET 9

H1

GE SERVICE
CD-ROM
Set Up & Cal:
➔ **LOADING/DELETING
PROPRIETARY
SOFTWARE**

Estimate **0.5 Hours**

Actual _____ Hours

Complete ☐

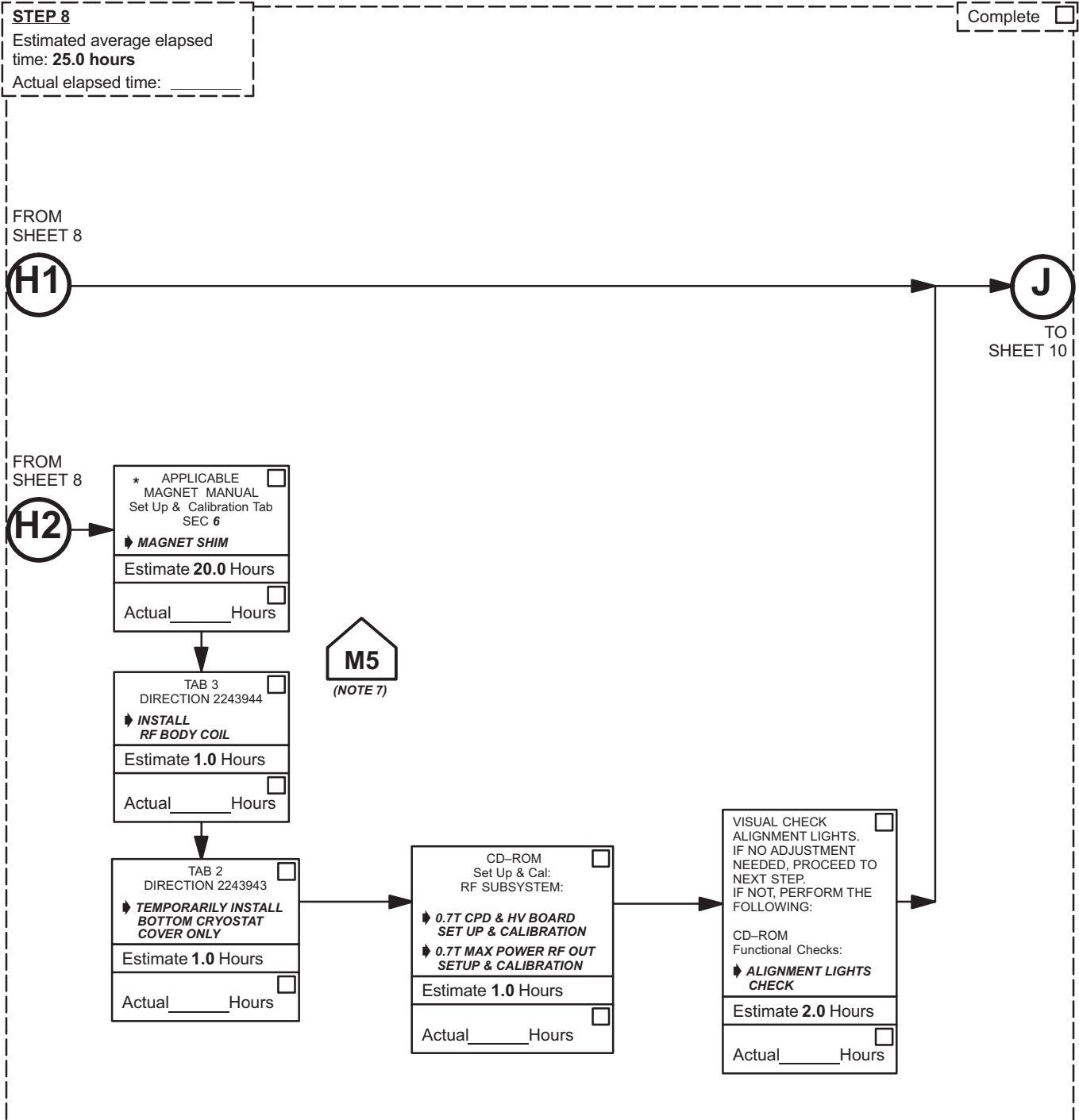
SIGNA OPENSPEED SYSTEM INSTALLATION FLOW CHART
ILLUSTRATION 1-5 (SHEET 8 OF 12)

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SYSTEM CALIBRATION (CONTINUED)



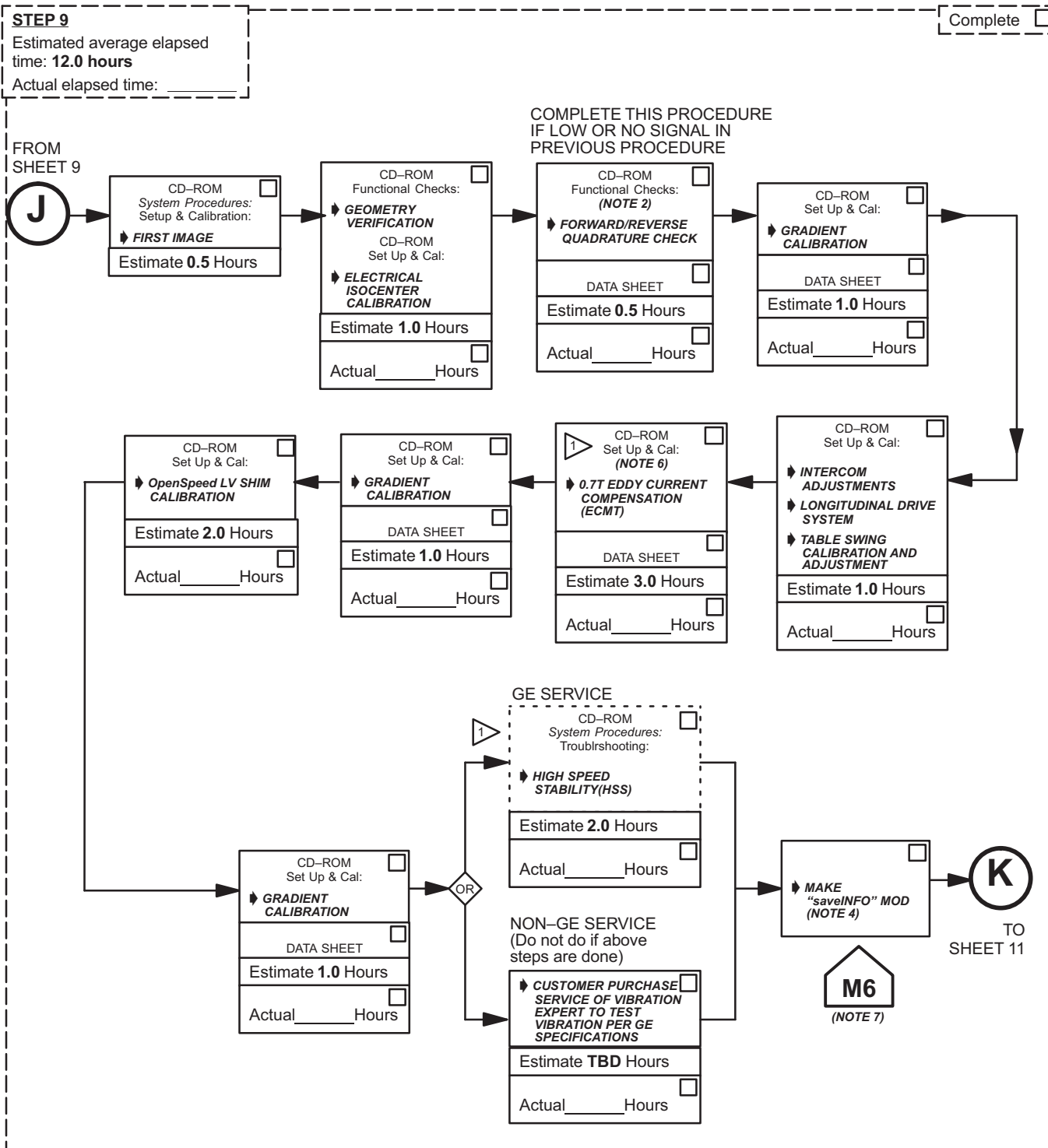
SIGNA OPENSPEED SYSTEM INSTALLATION FLOW CHART
 ILLUSTRATION 1-5 (SHEET 9 OF 12)

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SYSTEM CALIBRATION (CONTINUED)



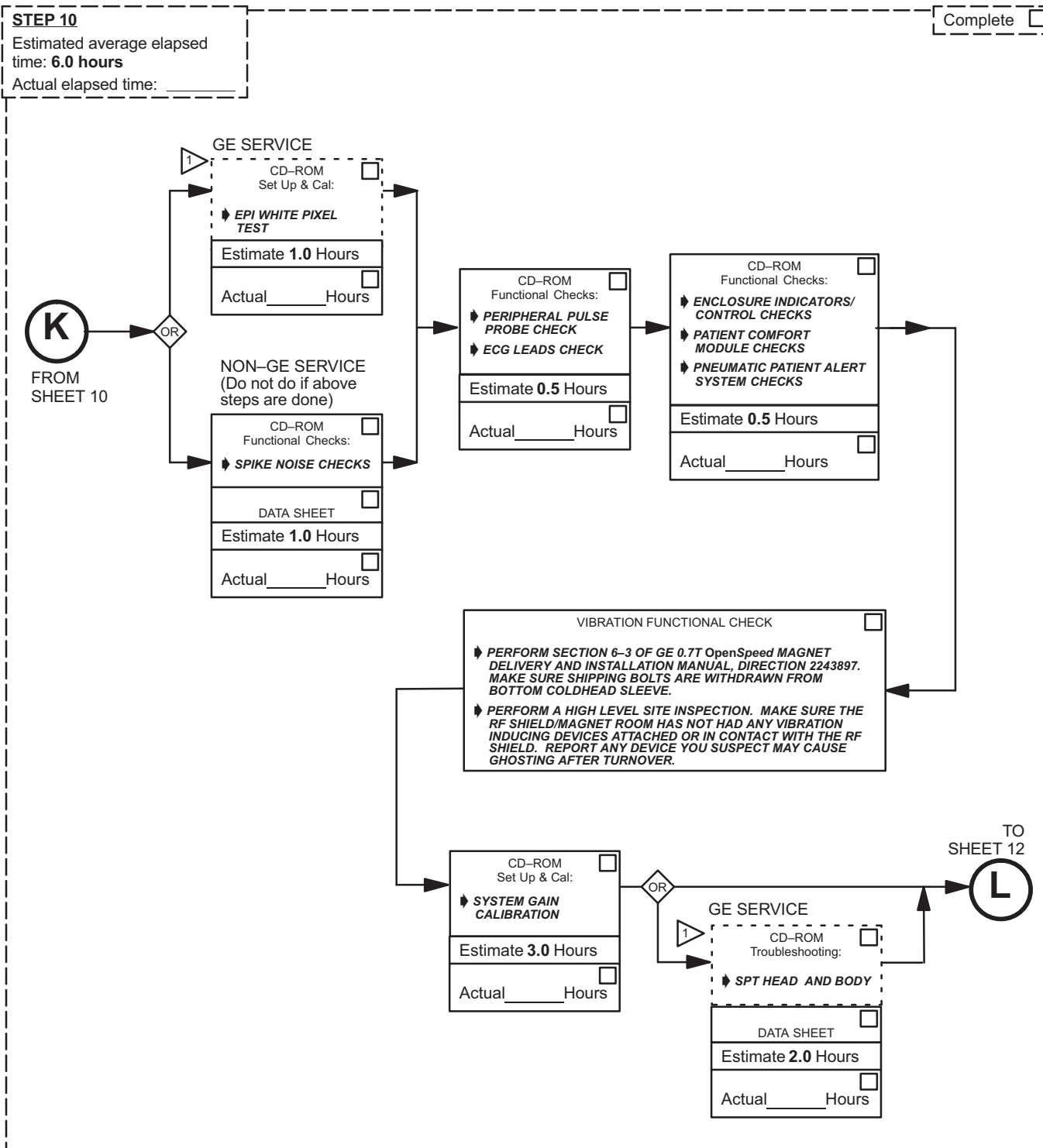
SIGNA OPENSPEED SYSTEM INSTALLATION FLOW CHART
 ILLUSTRATION 1-5 (SHEET 10 OF 12)

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SYSTEM CALIBRATION (CONTINUED)



SIGNA OPENSPEED SYSTEM INSTALLATION FLOW CHART
ILLUSTRATION 1-5 (SHEET 11 OF 12)

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IMPORTANT: PER ISO 9001 REQUIREMENTS

SYSTEM CALIBRATION (CONTINUED)

Complete ☐

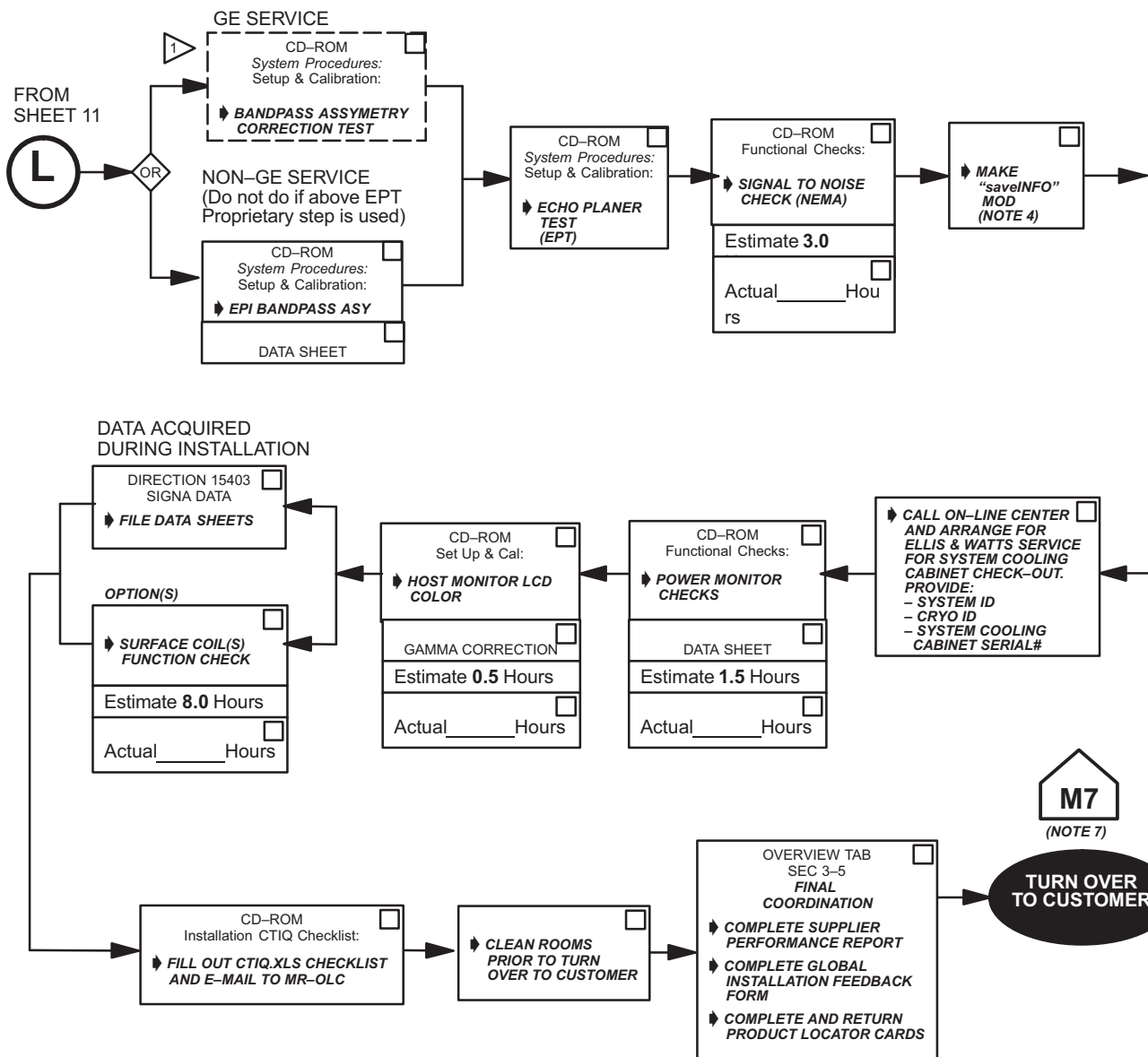
STEP 11

Estimated average elapsed time: **14.0 hours**

Actual elapsed time: _____

FOR EPI OPTION

(PERFORM PROCEDURES FOR HEAD)



SIGNA OPENSPEED SYSTEM INSTALLATION FLOW CHART
 ILLUSTRATION 1-5 (SHEET 12 OF 12)